FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2016**

PART TEST – IV

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 08)** contains 8 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 to 16) contains 4 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (17 – 20) contains 4 Matching Lists Type questions: Each question has four statements in LIST I & 4 statements in LIST II. The codes for lists have choices (A), (B), (C), (D) out of which only one is correct. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

This section contains **8 multiple** choice questions numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which ***ONE or MORE*** are correct.

1. The mass number of a nucleus is
   (A) May be equal to its atomic number.
   (B) Always less than its atomic number.
   (C) May be more than its atomic number
   (D) May be more than or may be equal to its atomic number.

2. The potential difference applied to an X-ray tube is increased. As a result in the emitted radiation.
   (A) The intensity increases. (B) The minimum wavelength increases.
   (C) The intensity remains unchanged. (D) The minimum wavelength decreases.

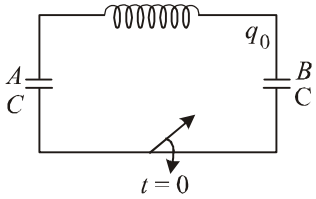
3. When photons of energy 4.0 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy K_A eV and de-Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.50 eV is $K_B = (K_A - 1.50)$ eV. If the de-Broglie wavelength of these photoelectrons is $\lambda_B = \sqrt{2}\,\lambda_A$, then
   (A) the work function of A is 1.0 eV (B) the work function of B is 3.0 eV
   (C) $K_A = 3.00$ eV (D) $K_B = 3.0$ eV

4. A prism of refractive index $\sqrt{2}$, refracting angle A produces minimum deviation D_m of a ray on one face at an angle of incidence 45° . The values of A and D_m are respectively.
   (A) 45°, 45° (B) 45°, 60°
   (C) 60°, 30° (D) 60°, 45°

Space for rough work

5. At t = 0 capacitor B has charge q_0

(Take : $\frac{1}{\sqrt{LC}} = \frac{100\pi}{\sqrt{2}}$, $q_0 = 10^{-3}C$)



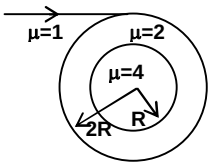
(A) If $i = i_{max}$ in the circuit then q will be same on each capacitor

(B) If $i = i_{max}$ in the circuit then $\frac{q_A}{q_B} = 2$

(C) $q_B = 50\,(1 - \cos 100\,\pi t) \times 10^{-3}$ C

(D) $q_A = 50\,(1 + \cos 100\,\pi t) \times 10^{-3}$ C

6. Figure shows two spherical surfaces of radii R and 2R separating three transparent media of refractive index $\mu = 4$, $\mu = 2$ and $\mu = 1$. A ray of light travelling in medium $\mu = 1$ is incident on outer sphere tangentially. The net deviation suffered by light ray will be.



(A) 0° (B) 90°

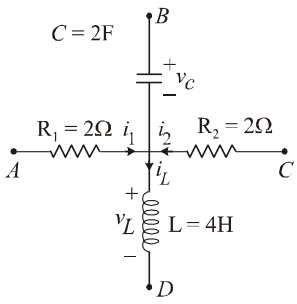
(C) 240° (D) 280°

7. The magnetic field within a long straight solenoid with a circular cross section and radius R increases at a rate of $\frac{dB}{dt}$. The magnitude of the induced electric field outside the solenoid at a distance r_2 from the axis is :

(A) $\frac{R^2}{r_2}\frac{dB}{dt}$ (B) $\frac{R^2}{2r_2}\frac{dB}{dt}$

(C) $\frac{3R^2}{2r_2}\frac{dB}{dt}$ (D) $\frac{R^2}{\sqrt{2}r_2}\frac{dB}{dt}$

8. In figure, $i_1 = 10e^{-2t}$, $i_2 = 4A$ and $v_c = 3e^{-2t}$ V. The current i_L is :



(A) $\left[2 - 2(1 - e^{-2t})\right]$A

(B) $\left[2 + 2(1 - e^{-2t})\right]$A

(C) $\left[3 - 2(1 - e^{-2t})\right]$A

(D) $\left[2 + 3(1 - e^{-2t})\right]$A

Space for rough work

Comprehension Type

This section contains **4 groups** of questions. Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 9 to 10

An electron is orbiting in a circular orbit of radius r under the influence of a constant magnetic field B,. instead of coulombic attraction. Assuming that Bohr's postulate regarding the quantization of angular momentum holds good for this electron [h = planks constant, e = charge on electron and m = mass of electron]

Answer the following questions in view of the above passage:

9. Radius of nth orbit of the electron will be

(A) $\sqrt{\frac{2nh}{\pi Be}}$ (B) $\sqrt{\frac{nh}{2\pi Be}}$

(C) $\sqrt{\frac{neh}{2\pi B}}$ (D) $\sqrt{\frac{2neh}{\pi B}}$

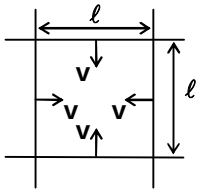
10. K.E of electron in n^{th} orbit will be:

(A) $\frac{nhBe}{4\pi m}$ (B) $\frac{nhBe}{2\pi m}$

(C) $\frac{nhB}{2\pi em}$ (D) $\frac{nhB}{4\pi em}$

Paragraph for Question Nos. 11 to 12

In the figure shown the figure rods have λ resistance per unit length. The arrangement is kept in a magnetic field of constant magnitude B and directed perpendicular to the plane of the figure and directed inwards. Initially the sides are given as shown and they form a square. Now each wire starts moving with constant velocity towards opposite wire.



11. Find as a function of time, induced emf in the circuit.

(A) $4vB(\ell - 2vt)$ (B) $8v^2Bt$

(C) $4v^2Bt$ (D) $4vB(\ell - 4vt)$

12. Induced current in the circuit with direction is

(A) $\frac{vB}{2\lambda}$, clockwise (B) $\frac{vB}{\lambda}$, clockwise

(C) $\frac{2vB}{\lambda}$, anticlockwise (D) none of these

Space for rough work

Paragraph for Question Nos. 13 to 14

A thin equi-convex glass lens with radius of each of face R is placed between water and oil.

(Given: $_a\mu_g = \frac{3}{2}$, $_a\mu_{oil} = 2$ and $_a\mu_w = \frac{4}{3}$)

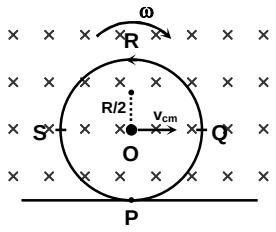
Answer the following questions in view of the above passage:

13. If incident ray is from oil then focal length will be :
(A) 4R (B) 2R
(C) (3/2)R (D) ∞

14. If incident ray is from water then focal length will be :
(A) 4R (B) 2R
(C) (3/2)R (D) 6R

Paragraph for Question Nos. 15 to 16

A conducting ring of radius R is rotating as well as translating on a smooth horizontal surface, where a uniform magnetic field B is present. Velocity of centre of mass is given as $v = 2R\omega$, where ω is the angular velocity of the ring, then



15. Motional emf across PR is
(A) 2BRv (B) BRv
(C) 3/4BRv (D) 4BRv

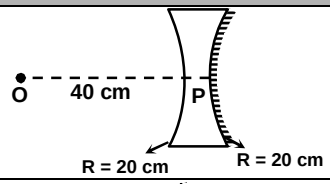
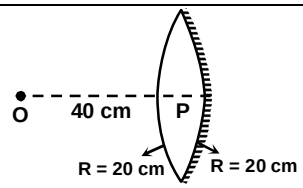
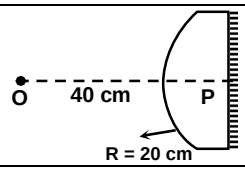
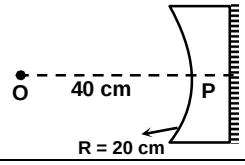
16 Motional emf across PS is
(A) 2BRv (B) BRv
(C) 3/4BRv (D) 4BRv

Space for rough work

Matching List Type

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Refractive index of lenses is 3/2. Then the final position of the image from the pole is

List – I		List - II	
(P)		(1)	40/7 cm, left to P
(Q)		(2)	40/9 cm, right to P
(R)		(3)	40/3 cm, right to P
(S)		(4)	40 cm, left to P

Codes:

	P	Q	R	S
(A)	4	1	2	1
(B)	2	4	1	3
(C)	4	3	1	2
(D)	2	1	4	3

18. Match the following list-I with List-II

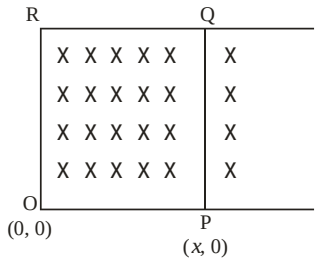
List - I		List - II	
(P)	Wave nature of light	(1)	Young's double slit experiment
(Q)	Particle nature of light	(2)	Photoelectric effect
(R)	Wave nature of electrons	(3)	Electron microscope
(S)	Particle nature of electrons	(4)	Cathode rays

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	2	4	1	3
(C)	4	3	1	2
(D)	2	3	4	1

Space for rough work

19. Consider two parallel conducting frictionless, resistanceless tracks which are kept in gravity free space as shown in the figure. A moveable conductor PQ of resistance r initially kept at OR given a velocity v_0 m/s towards right. If there exists a magnetic field which depends on x-co-ordinate in space from O and perpendicular to plane of parallel tracks and given by



$\vec{B} = \alpha x(-\hat{k})$ (α is constant)

Take mass of conductor PQ m = 1 kg, length l = 1m α = 1 T/m, v_0 =10 m/s. ($2^{1/3}$ = 1.26)

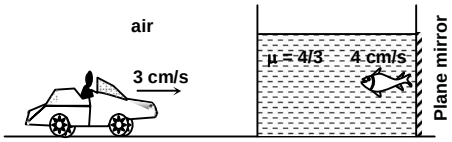
r = $\frac{9}{5}\Omega$. Then match List-I with List-II.

List – I		List - II	
(P)	x-coordinate of conductor when speed become half of initial value.	(1)	15 unit
(Q)	Current of conductor when speed become half of initial value	(2)	3 unit
(R)	E.M.F. induced in conductor when speed become half of initial value.	(3)	4 unit
(S)	x-coordinate when speed becomes zero (approximately)	(4)	$\frac{25}{3}$ unit

Codes:

	P	Q	R	S
(A)	1	4	2	1
(B)	4	2	1	3
(C)	2	4	1	3
(D)	2	3	4	1

20. A child in a toy car is moving towards a cubical container with a velocity of 3 cm/s, whereas fish is moving towards child with a velocity of 4 cm/s as shown in the figure. The wall behind the fish is silvered (Refractive index of water is μ = 4/3). Match the following



List – I		List - II	
(P)	Velocity of child as observed by fish by one refraction only	(1)	zero
(Q)	Velocity of fish as observed by child by one refraction only	(2)	6 cm/s
(R)	Velocity of image of fish by reflection from mirror as observed by child.	(3)	8 cm/s
(S)	Velocity of image of child by reflection from mirror as observed by fish	(4)	7 cm/s

Codes:

	P	Q	R	S
(A)	1	4	2	1
(B)	3	2	1	1
(C)	2	4	1	3
(D)	2	3	4	1

Space for rough work

Chemistry

PART – II

SECTION – A
Multiple Correct Answers Type

*This section contains 8 multiple correct answer(s) type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE** is/are correct.*

1. Cataphoresis is the process used for the coagulation of colloids. It can be used for
   (A) colloids having positive charge (B) colloids having negative charge
   (C) lyophilic colloids (D) All of the above

2. The solid compound SiO_2 exist in the forms of
   (A) crystalline in quartz (B) amorphous in quartz
   (C) amorphous in silica glass (D) crystalline in silica glass

3. Given $E^o_{Ag^+|Ag} = 0.80V, E^o_{Mg^{2+}|Mg} = -2.37V, E^o_{Cu^{2+}|Cu} = 0.34V, E^o_{Hg^{2+}|Hg} = 0.79V$. Which of the following statement(s) is/are incorrect?
   (A) $AgNO_3$ can be stored in copper vessel.
   (B) $Cu(NO_3)_2$ can be stored in magnesium vessel.
   (C) $CuCl_2$ can be stored in silver vessel.
   (D) $HgCl_2$ can be stored in copper vessel.

4. Identify the correct statement(s)
   (A) 1 M NaCl has higher freezing point than 1 M glucose solution.
   (B) 1 M NaCl solution has almost same boiling point as 1 M KCl solution, assuming 100% ionisation.
   (C) Apparent molecular weight of NaCl will be less than 58.5 in water because it undergoes dissociation.
   (D) $i = 1$ when acetic acid is dissolved in benzene.

5. Which of the following is/are true?
   (A) All halogens exist as diatomic gaseous molecules at ordinary temperature
   (B) Bromine dissolves in chloroform imparting blue colour
   (C) Chlorine monoxide is evolved when $KClO_3$ is treated with conc. H_2SO_4
   (D) Chloride ion is confirmed by chromyl chloride test

Space for rough work

6. Which of the following is/are correct for lyophillic aquasols?
   (A) Its surface tension is lower than that of H_2O
   (B) Its viscosity is higher than that of water
   (C) Colloidal silver
   (D) Colloidal antimony

7. Which of the following compounds are coloured due to charge transfer spectrum?
   (A) K_2CrO_7 (B) $KMnO_4$
   (C) $[Co(NH_3)_6]Cl_3$ (D) $[Cu(NH_3)_4]SO_4$

8. Nitrogen dioxide can be obtained by heating
   (A) KNO_3 (B) $Pb(NO_3)_2$
   (C) $LiNO_3$ (D) $AgNO_3$

Linked Comprehension Type

*This section contains 4 paragraphs. Based upon each paragraph, 2 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 9 to 10

$(A) + NaCl \longrightarrow B(\text{white ppt.})$

$(B) + KI \longrightarrow C(\text{green ppt.})$

$(C) + KI(\text{excess}) \longrightarrow (D) + (E)(\text{colourless solution})$

$(E) + NH_3 + KOH \longrightarrow F$

9. Compound (A) and (B) are respectively
   (A) $AgNO_3$ and AgCl (B) $Pb(NO_3)_2$ ad $PbCl_2$
   (C) $Hg_2(NO_3)_2$ and Hg_2Cl_2 (D) $Cu_2(NO_3)_2$ and Cu_2Cl_2

10. When compound (B) reacts with NH_3 solution, the colour of the compound formed is
   (A) black (B) red
   (C) yellow (D) white

Space for rough work

Paragraph for Question Nos. 11 to 12

The binary compounds of oxygen with other elements are called oxides. They are classified either depending upon their acid – base characteristics or on the basis of oxygen content.

11. Which of the following pairs contain neutral oxides?
(A) SO_2, SO_3 (B) N_2O_3, N_2O_5
(C) CO, NO (D) Na_2O, CaO

12. Which of the following pairs contains mixed oxides?
(A) Pb_3O_4, Fe_3O_4 (B) MnO_2, BaO_2
(C) KO_2, Na_2O_2 (D) Mn_3O_4, N_2O_5

Paragraph for question Nos 13 to 14

The electrical energy produced by a galvanic cell is given by the product of its electromotive force and the quantity of electricity which passes. Till about the middle of the nineteenth century it was generally believed that electrical energy of a reversible cell originated from the decrease in the enthalpy ($-\Delta H$) of the cell reaction. This view received support from the fact that the enthalpy change of the reaction

$$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$$

taking place in the Daniell cell is about 210,000 joules which is very close to the electrical energy of the cell viz., 212,300 joules, as calculated above. However, towards the close of the century, it was pointed out by Gibbs, and independently by Helmholtz, that the electrical energy of a reversible cell originates from the free energy decrease ($-\Delta G$) of reaction occurring in the cell. Suppose in a particular cell reaction, n is the number of electrons liberated at one electrode (or taken up at the other electrode), then evidently, n faradays (nF coulombs) of electricity will be generated in the complete cell reaction. If, for the sake of simplicity, the EMF of the cell, viz., E_{cell} is denoted by E, then

Electrical energy produced by the cell = nFE

Hence, $-\Delta G = nFE$

Read the following paragraph and answer the questions given below:

13. In the Daniell cell, the EMF is 1.10 volts. The cell reaction involves liberation as well as taking up of 2 moles of electrons. What will be the electrical energy generated in the cell for the complete cell reaction
(A)106, 150 joule (B) 212, 300 joule
(C) 53, 075 joule (D) None of these

14. For the Daniel cell involving the cell reaction,
$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$
The standard free energies of formation of Zn(s), Cu(s), $Cu^{2+}(aq)$ and $Zn^{2+}(aq)$ are 0, 0, 64.4 kJ mol^{-1} and –154.0 kJ mol^{-1}, respectively. Calculate the standard EMF of the cell.
(A) 1.13 volt (B) 2.10 volt
(C) 0.5 volt (D) 1.95 volt

Space for rough work

Paragraph for question Nos 15 to 16

Addition of non-volatile solute always lowers the vapour pressure of solvent, therefore, it will be in equilibrium with solid phase at a lower pressure and hence at a lower temperature, the difference between the freezing points of the pure solvent and its solution is called depression of freezing point. The depression in freezing point when 1 mol of solute is dissolved in 1000 gm of solvent is called molal freezing point depression constant (K_f). The value of K_f for H_2O is 1.86 K mol^{-1}kg. To understand this, a solution of 25% (w/w) $MgCl_2$ containing impurity of the chloride of a trivalent metal M having atomic mass 43.5 gm (24% by weight of $MgCl_2$) has been prepared. In this solution $MgCl_2$ is completely dissociated. On the other hand only 50% dissociation of MCl_3 takes place and molality is approximately equal to molarity.

Read the following paragraph and answer the questions given below:

15. Depression in freezing point of the solution is
(A) 17.36 K (B) 13.02 K
(C) 14.88 K (D) 12.42 K

16. If the solution is cooled to –21.7°C, what weight of ice will be formed?
(A) 15 gm (B) 60 gm
(C) 40 gm (D) 51 gm

(Matrix-Match Type)

*This section contains **4 questions.** Each question has four statements (P, Q, R and S) given in **List-I** and **four statements** (1, 2, 3, 4) in **List-II.** Any given statement in **List**-I can have correct matching with **ONE** or **MORE** statement(s) given in **List**-**II**. For example, if for a given question, statement B matches with the statements given q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.*

17. Match the List – I with the List - II.

List-I		List-II	
(P)	$\wedge_m$	(1)	$\frac{1}{R}\times\frac{1}{\alpha}$
(Q)	κ (kappa)	(2)	$\frac{1000\times\kappa}{M}$
(R)	α	(3)	$\wedge^{\circ}_{NaCl}=\lambda^{\circ}_{Na^+}+\lambda^{\circ}_{Cl^-}$
(S)	Kohlrausch's law	(4)	$\frac{\wedge}{\wedge^{\circ}}$

Codes:	**P**	**Q**	**R**	**S**
(A)	2	1	4	3
(B)	2	3	4	1
(C)	4	1	2	3
(D)	1	4	2	3

Space for rough work

18. Match the List-I with the List-II

List-I	List-II
(P) van Arkel method	(1) Purification of copper
(Q) Poling	(2) Refining of silver
(R) Cupellation	(3) Manufacture of Na_2CO_3
(S) Solvay process	(4) Purification of titanium

Codes:	**P**	**Q**	**R**	**S**
(A)	3	2	1	4
(B)	2	3	4	1
(C)	4	1	3	2
(D)	4	1	2	3

19. Match the List – I with List – II:

List-I		List-II	
(P)	Molality of water	(1)	1.13
(Q)	Mole fraction of sugar in 214.2 g sugar syrup containing 34.2 g of sugar.	(2)	0.555
(R)	Molality of sugar solution(II) prepared above	(3)	0.0099
(S)	Molality of CH_3OH is aqueous CH_3OH solution (Mole fraction of CH_3OH = 0.02 and density of solution = 0.994 g/cc)	(4)	55.5

Codes:	**P**	**Q**	**R**	**S**
(A)	3	2	1	4
(B)	4	3	2	1
(C)	4	1	2	3
(D)	1	4	2	3

20. Match the geometry (List – I) with the corresponding complex (List – II).

List – I (Geometry)		List – II (Complex)	
(P)	Octahedral	(1)	Tetracarbonyl nickel (0)
(Q)	Square planar	(2)	Pentafluoro antimonate (III) ion
(R)	Tetrahedral	(3)	Tetraammine copper (II) sulphate
(S)	Square pyramidal	(4)	Hexacyanoferrate (III) ion

Codes:	**P**	**Q**	**R**	**S**
(A)	3	2	1	4
(B)	2	3	4	1
(C)	4	3	1	2
(D)	1	4	2	3

Space for rough work

Mathematics PART – III

SECTION – A

Multiple Correct Choice Type

*This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct*

1. The line whose vector equations are $\vec{r} = 2i - 3j + 7k + \lambda(2i + pj + 5k)$ and $\vec{r} = i + 2j + 3k + \mu(3i - pj + pk)$ are perpendicular for all values of λ and μ if p is equal to
   (A) –1 (B) 2
   (C) 5 (D) 6

2. The equation of projection forming plane of the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{4}$ in the plane x + 2y + z = 6 is/are
   (A) x + 2y + z – 6 = 0 (B) 9x – 2y – 5z + 4 = 0
   (C) x – 2y – z + 6 = 0 (D) 9x + y – 5z + 4 = 0

3. An urn contains four tickets with numbers 112, 121, 211, 222 and one ticket is drawn. Let Ai(i = 1, 2, 3) be the event that the ith digit of the number of ticket drawn is 1 then
   (A) $P(A_1) = P(A_2) = P(A_3) = 1/2$
   (B) $P(A_1 \cap A_2) = P(A_1) \cdot P(A_2)$
   (C) A_1, A_2, A_3 are pairwise independent events
   (D) $P(A_1 \cap A_2 \cap A_3) \neq P(A_1)\, P(A_2) P(A_3)$

4. Let us define the events A and B as
   A : An year chosen at random contains 29 days in the February month.
   B : An year chosen at random has does not contain 53 Fridays.
   If P(E) denotes the probability of happening of event E then
   (A) $P(\bar{B})=\frac{2}{7}$ (B) $P(B) = \frac{23}{28}$
   (C) $P(A/\bar{B}) = 2/5$ (D) P(A/(B) = 5/23

Space for rough work

5. If A and B are any two different square matrices of order n with A – B is non-singular $A^3 = B^3$ and A(A(B)) = B(B(A)), then

(A) $A^2 + B^2 = O$ (B) $A^2 + B^2 = I$

(C) $A^2 + B^3 = I$ (D) $A^3 + B^3 = O$

6. Let $\Delta = \begin{vmatrix} a & a+d & a+2d \\ a+d & a+2d & a \\ a+2d & a & a+d \end{vmatrix}$, then

(A) Δ depends on a (B) Δ depends on d

(C) Δ is constant (D) all the above

7. Area bounded by curve $y^2 = x$ and $x = 4$ is divided into 4 equal parts by the lines $x = a$ and $y = b$ then.

(A) Area of each part = $\frac{8}{3}$ (B) $b = 0$

(C) $a = \sqrt{2}$ (D) $a = (16)^{1/3}$

8. The differential equation of the curve for which the initial ordinate of any tangent is equal to the corresponding subnormal.

(A) is linear in x (B) is homogenous of first degree

(C) has separable variables (D) is of second order.

Space for rough work

Comprehension Type

*This section contains **4 groups of questions**. Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct*

Paragraph for Question Nos. 9 to 10

$|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{a} + \vec{b}| = 1$, $\vec{a} \cdot \vec{c} = 0$. If $\vec{a} = \frac{\hat{i}}{\sqrt{2}} + \frac{\hat{j}}{\sqrt{2}}$ and $\vec{a}$, $\hat{k}$ and $\vec{b}$ are linearly dependent.

9. If maximum and minimum value of $\vec{b} \cdot \vec{c}$ is M and m, then $4(M^2 + m^2)$ is
   (A) 1 (B) 5
   (C) 3 (D) 6

10. If for some $\vec{c}$, $|\vec{b} \times \vec{c}| = |\vec{a} \times \vec{c}|$ and $\vec{c} = p\hat{i} + q\hat{j} + r\hat{k}$, then $16(p^4 + q^4 + r^4)$ is
   (A) 8 (B) 21
   (C) $\frac{21}{3}$ (D) 4

Paragraph for Question Nos. 11 to 12

T is the region of the plane x + y + z = 1 with x, y, z > 0. S is the set of points (a, b, c) in T such that just two of the following three inequalities hold: $a \le \frac{1}{2}$, $b \le \frac{1}{3}$, $c \le \frac{1}{6}$.

11. Area of the region T is
   (A) $\frac{\sqrt{3}}{4}$ (B) $\frac{\sqrt{3}}{2}$
   (C) $\sqrt{3}$ (D) none of these

12. Area of the region S is
   (A) $\frac{\sqrt{3}}{72}$ (B) $\frac{7\sqrt{3}}{36}$
   (C) $\frac{\sqrt{3}}{4}$ (D) none of these

Space for rough work

Paragraph for Question Nos. 13 to 14

A matrix H = $[h_{jk}]$ is used to keep track of three football players (numbered 1, 2, and 3) in three matches (1st, 2nd and 3rd).
$Re(h_{jk})$ = number of matches in which both j^{th} and k^{th} players played or both did not play.
(If j = k it is 3)
$Img(h_{jk})$ = (number of matches played by k^{th} player) – (number of matches played by j^{th} player)

13. Let P = $[P_{jk}]$, P_{jk} = 1 if j^{th} player played in k^{th} match and P_{jk} = i otherwise $\left(i = \sqrt{-1}\right)$, then |det(H)| is equal to
(A) |det(P)| (B) $|det(P)|^2$
(C) $|det(P)|^3$ (D) none of these

14. det(H) is
(A) complex with imaginary part a multiple of 4 (B) non–positive multiple of 3
(C) non negative multiple of 4 (D) none of these

Paragraph for Question Nos. 15 to 16

Given $f(x) = x^3 + x - 2$, $f: R \to R$, area of bounded by f(x), $f^{-1}(x)$ and xy = 0 in 1st quadrant is A, where $A = 3t - \frac{5}{2}$, where $t = a^{1/b}$ and 3(a + b) = n; a, b, n $\in$ N and a is prime.

15. Difference of last two digits of $a^{a^{11}-36}$ is
(A) 0 (B) 1
(C) 2 (D) 3

16. Number of ways of putting n alike apples and n alike oranges in six baskets if each basket can hold atmost five fruits is
(A) $^{20}C_5 - 6\ ^{14}C_5 + 15\ ^8C_5$ (B) $(^{20}C_5 - 6\ ^{14}C_5 + 15\ ^8C_5)^2$
(C) $(^{20}C_5 - 6\ ^{14}C_5 + 15\ ^8C_5)^2 - 5^6$ (D) none of these

Space for rough work

(Matching list Type)

*This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct*

17. Now match the following List–I with List–II.

	List – I		**List – II**
(P)	$\vec{a}, \vec{b}$ unit vectors and $\vec{a}+2\vec{b} \perp 5\vec{a}-4\vec{b}$, then $2(\vec{a}\cdot\vec{b})$ is equal to	(1)	0
(Q)	The points (1, 0, 3), (–1, 3, 4), (1, 2, 1), (k, 2, 5) are coplanar, then k =	(2)	-1
(R)	The vectors (1, 1, m), (1, 1, m + 1), (1, –1, m) are coplanar, then number of values of m is	(3)	1
(S)	$\vec{a}\times(\vec{b}\times\vec{c})+\vec{b}\times(\vec{c}\times\vec{a})+\vec{c}\times(\vec{a}\times\vec{b})$ is equal to	(4)	2

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	3	2	1	1
(C)	3	2	4	1
(D)	4	1	2	3

Space for rough work

18. India and Australia play a series of 'n' one day matches and probability that India wins a match against Australia is $\frac{1}{2}$.

List – I		List – II	
(P)	If 'n' is not fixed and series ends when any one of the team completes its 4th win then probability that India wins the series is	(1)	$\frac{4}{2^7}$
(Q)	If n = 7; then probability that India wins atleast three consecutive matches is	(2)	$\frac{47}{2^7}$
(R)	For n = 7, probability that India wins series through consecutive wins is	(3)	$\frac{1}{2}$
(S)	For n = 7, probability that India doesn't win two consecutive matches is	(4)	$\frac{17}{2^6}$

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	3	2	4	1
(C)	3	2	1	4
(D)	4	1	2	3

Space for rough work

19. Consider the system of equations:
$a^2x + (a^2 – (b – c)^2)y + bcz = 0$
$b^2x + (b^2 – (c – a)^2)y + caz = 0$
$c^2x + (c^2 – (a – b)^2)y + abz = 0$, then

List – I		List – II	
(P)	If a + b + c = 0 and no two out of a, b, c are equal then	(1)	system of equations have unique solution
(Q)	If a = b = c = 0, then	(2)	system of equations have infinite number of solutions
(R)	If a = b = c, a + b + c $\neq$ 0, then	(3)	$\forall$ x, y, z $\in$ R, the system of equations will be true
(S)	If a + b + c $\neq$ 0 and no two out of a, b, c are equal, then	(4)	solutions of system of equations are in the form $(\lambda_1, \lambda_2, -\lambda_1 - \lambda_2)$, where $\lambda_1, \lambda_2 \in R$

Codes:

	P	Q	R	S
(A)	2	2, 4	2, 3	3
(B)	4	3, 2	3	4
(C)	4	3, 4	4	1
(D)	2	2, 3	2, 4	1

Space for rough work

20. Now match the following List–I with List–II.

List – I		List – II	
(P)	Solution of the differential equation $y'' + e^{2y}(y')^3 = 0$ is	(1)	$x = \frac{1}{4}e^{2y} + c_1y + c_2$
(Q)	Solution of the differential equation $2x(x + y)y' = 3y^2 + 4xy$ at $y(1) = 1$ is	(2)	$y^2 + 2xy = 3x^3$
(R)	Equation of orthogonal trajectories of the curve $y = \tan x + k$ is	(3)	$2x + 4y + \sin 2x = c$
(S)	Solution of the differential equation $y' + 4xy + xy^3 = 0$ is	(4)	$y^2 = \left(ce^{4x^2} - \frac{1}{4}\right)^{-1}$

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	1	2	3	4
(C)	1	2	4	3
(D)	4	1	2	3

Space for rough work